Laws of exponents

1 Simplify the following expressions:

a $a^s \times a^t$ b $\dfrac{a^s}{a^t}$ c $a^s \times a^s$ d $a^s + a^s$

e $a^s + a^t$ f $2^2 \times 2^2$ g $4y^3 \times 6y$ h $3x^5 \times 8x^2$

2 Simplify the following expressions:

a $\dfrac{x^6}{4x^4}$

b $p^{18} \div p^8 \div p^5$

c $\dfrac{4n^3 \times 4n^4}{16}$

d $\dfrac{u^{2x+1}}{u^x}$

e $p^7 \div p^3 \times p^5$

f $p^{14} \div p^6 \div p^5$

g $\dfrac{y^7 \times y^5}{y^3 \times y^2}$

h $12s^8 \div 3t^6 \times 6t^4$

i $\dfrac{10p^6 \times 3p^{10}}{15p^2}$

j $c^{2z}c^{z+1}$

k $20m^6 \div 5m^{13} \times 9m^2$

l $\dfrac{12y^4 \times 35y^8}{4y^5 \times 5y^4}$

3 Complete the statement below:

$$15j^{14} \div (\square) = 5j^7$$

4 Rewrite the following expressions without parantheses:

a $\left(\dfrac{a}{b}\right)^3$ b $(xy)^7$ c $(2x)^4$ d $\left(\dfrac{xy}{5z}\right)^2$

5 Write $(16^p)^4$ in the form a^b, where a is a prime number.

6 For each of the following expressions, state the operation that should be done to the exponents to simplify the expressions:

a $\left(z^{28}\right)^{19}$ b $z^{13} \times z^{16}$ c $z^{14} \div z^{13}$ d $\dfrac{z^{26}}{z^{13}}$

7 Simplify the following expressions:

a $\frac{(x^3)^2}{x^3}$

b $(u^{x+1})^3$

c $p^2 \times (p^3)^2$

d $p^8 \div (p^2)^2$

e $\frac{(x^2)^6}{(x^2)^2}$

f $(2y^4)^2 \times (2y^2)^3$

g $\frac{(4y^3)^5}{(y^2)^2}$

h $(3y^2)^3 \times (-2y^2)^2$

i $(u^9 \times u^5 \div u^{19})^2$

j $\frac{28m^4 \times (3m^3)^3}{7m^2 \times (m^2)^2}$

k $\frac{(-35m^4) \times (3m^4)^3}{7m^3 \times (m^2)^5}$

l $\left(\frac{t^4}{t^2}\right)^5 \div \frac{t^5}{t^8}$

m $(t^6)^5 \div \left(\frac{t^9}{t^7}\right)^3$

n $\frac{(u^{x+3})^3}{u^{x+1}}$

Zero and negative exponents

8 Simplify:

a a^0

b $8x^0$

c $(2 \times 13)^0$

d $(a^0)^{79}$

e $(13x^7)^0 + 13^0 - 13h^0$

f $(7m^0 + 4)^2$

9 Express the fraction $\frac{m}{n^4}$ as a product using negative exponents.

10 Express the following in simplest form without negative exponents:

a $4a^{-2}$

b $(3a)^{-1}$

c $6x^{-3}y^4$

d $m^2 \times m^{-7}$

e $m^{-5}n^{-4}p^4$

f $2y^9 \times 3y^{-5}$

g $\frac{9x^3}{3x^{-4}}$

h $\frac{12x^3}{4x^7}$

11 Express the following in simplest form with[redacted]gative exponents:

a $\left(\dfrac{a}{b}\right)^{-5}$ b $\left(4m^{-10}\right)^{4}$ c $\left(4m^{-8}\right)^{-3}$ d $\left(5y^{3}\right)^{-3}$

e $\left(\dfrac{x^2}{y^4}\right)^{-1}$ f $\left(\dfrac{a^3}{b^3}\right)^{-5}$ g $\left(\dfrac{2h}{3}\right)^{-4}$ h $\dfrac{\left(m^{-3}\right)^{-1} \times \left(m^{4}\right)^{-3}}{m^3 \times m^4}$

12 Express the following in simplest form without negative exponents:

a $2p^4q^{-2} \times 5p^{-4}q^{-5}$ b $\dfrac{5p^5q^{-4}}{40p^5q^6}$

c $10x^3y^2z \div 2x^5y^9$ d $2^{15} \div 2^{-5}$